UNIT 1 · STANDARD 6.NS.3

Divide Decimals

Lesson 1-7

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In 8.4 ÷ 2.1 = 4, the dividend is 8.4 — it is the total being split

Dividend

The number you are splitting up in a division problem.

In 8.4 ÷ 2.1 = 4, the divisor is 2.1 — it is the size of each group

Divisor

The number you split by in a division problem.

In 8.4 ÷ 2.1 = 4, the quotient is 4 — there are 4 groups

Quotient

The answer when you divide.

8.4 ÷ 2.1 → multiply both by 10 → 84 ÷ 21 = 4

Decimal division

Dividing with decimals. First make the number you divide by a whole number.

7.56 ÷ 0.12 = 756 ÷ 12 = 63 (both multiplied by 100)

Equivalent division

Multiplying both numbers by 10 or 100 gives the same answer.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can divide with decimals by making the divisor a whole number first.

1️⃣ Notice

The space station has 18.9 kilograms of a rare mineral sample that must be divided equally among 6.3-kilogram storage pods.

2️⃣ Key idea

I can divide with decimals by making the divisor a whole number first.

3️⃣ Words I'll use

Dividend **Divisor**

Quotient

Decimal division

Equivalent division

🤔 Think About It

- What total amount is being divided?
- What is the size of each pod?
- How do you divide when the divisor is a decimal?

👀 Watch

Watch your teacher model one example. Jot what you see.

____

🤝 We try

Solve the next one together as a class.

____

____

✏️ You try

Now try one on your own.

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The station has 18.9 kilograms of mineral to divide into 6.3-kilogram pods. Why can't you divide by the decimal 6.3 directly the same way you divide by a whole number?

Level 1 support

Start here: You first make the divisor a whole number using equivalent division.

💡 **Need a hint?**

Sentence starters (optional)

I make the divisor 6.3 into a ____ before dividing.

Convierto el divisor 6.3 en un ____ antes de dividir.

Dividing by a decimal directly is hard because ____.

Dividir entre un decimal directamente es difícil porque ____.

Word bank: dividend | divisor | quotient | decimal division | equivalent division

Listen for: *Students explain you move the decimal in the divisor to make it whole (6.3 to 63) and move the dividend the same way, naming 18.9 as the dividend and 6.3 as the divisor.*

Level 2

Estimate how many pods can be filled before computing, then justify. About how many 6-kg groups fit in 18.9 kg?

- I estimate about ____ pods because 18.9 is close to ____.
- Dividing roughly 18 by 6 gives ____, so ____.

EXPLORE

When you divided 18.9 by 6.3, how did you use equivalent division to make the divisor a whole number?

Level 1 support

Start here: Move both decimal points the same number of places so the value stays equal.

💡 **Need a hint?**

Sentence starters (optional)

I moved both decimal points ____ place, changing it to ____ ÷ ____.

Moví ambos puntos decimales ____ lugar, cambiándolo a ____ ÷ ____.

The quotient is ____.

El cociente es ____.

Word bank: dividend | divisor | quotient | decimal division | equivalent division

Listen for: *Students rewrite 18.9 ÷ 6.3 as 189 ÷ 63 = 3, explaining moving both points one place keeps the division equivalent, so 3 pods are filled.*

Level 2

Why does moving BOTH decimal points the same number of places keep the answer the same? Justify with what you know about equivalent fractions.

- Moving both points keeps the answer the same because ____.
- It is like multiplying the top and bottom of a fraction by ____, which ____.

CONNECT

A ribbon is 16.8 feet long and you cut pieces that are each 2.4 feet. How does decimal division tell you how many pieces you can cut?

Level 1 support

Start here: Use equivalent division to make 2.4 whole, then divide.

💡 **Need a hint?**

Sentence starters (optional)

I can cut ____ pieces because I divided 16.8 by ____.

Puedo cortar ____ piezas porque dividí 16.8 entre ____.

I moved both decimals to change it to ____ ÷ ____.

Moví ambos decimales para cambiarlo a ____ ÷ ____.

Word bank: **dividend** **divisor** **quotient** **decimal division** **equivalent division**

Listen for: *Students rewrite 16.8 ÷ 2.4 as 168 ÷ 24 = 7, so 7 pieces can be cut with none left over.*

Level 2

How can you check that 7 pieces is correct without redividing? Show the check.

- I can check by multiplying ____ × ____ = ____.
- If the product equals the ribbon length, then ____.

REFLECT

Why does making the divisor a whole number first make dividing by a decimal so much easier?

Level 1 support

Start here: Equivalent division turns a decimal divisor into a familiar whole-number problem.

💡 **Need a hint?**

Sentence starters (optional)

Making the divisor a whole number helps because ___.

Convertir el divisor en un número entero ayuda porque ___.

After moving the points, the problem becomes a ___ division.

Después de mover los puntos, el problema se vuelve una división de ___.

Word bank: **dividend** **divisor** **quotient** **decimal division** **equivalent division**

Listen for: *Students explain that a whole-number divisor lets them divide normally, and equivalent division keeps the quotient the same while removing the decimal from the divisor.*

Level 2

In 19.2 ÷ 0.8, how many places do you move each decimal point, and why does the answer get larger than 19.2? Explain.

- I move each point ___ place, making it ___ ÷ ___.
- The quotient is larger than 19.2 because dividing by less than 1 ___.

Guided Examples

Example 1

What is 14.4 ÷ 1.2?

Solution: Multiply both by 10: 144 ÷ 12 = 12.

Answer: A. 12

Example 2

What is 3.78 ÷ 0.6?

Solution: Multiply both by 10: 37.8 ÷ 6 = 6.3.

Answer: A. 6.3

Example 3

What is 8.4 ÷ 0.7?

Solution: 8.4 ÷ 0.7 = 84 ÷ 7 = 12.

Answer: A. 12

Write About the Math

The Writing Revolution

I can explain my work using the words dividend, divisor, quotient, and equivalent division.

1. Kernel Sentence

subject + verb

Model: Decimal division is dividing with decimals. First make the number you divide by a whole number.
División decimal es dividir con decimales. Primero haz entero el número entre el que divides.

Write a kernel sentence about decimal division. Use a subject and a verb.
Escribe una oración base sobre división decimal. Usa un sujeto y un verbo.

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2. Sentence Expansion

because · but · so

Kernel: Decimal division matters in math
División decimal importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Decimal division matters in math because ____.** *División decimal importa en matemáticas porque ____.*
but *pero*	**Decimal division matters in math, but ____.** *División decimal importa en matemáticas, pero ____.*
so *entonces*	**Decimal division matters in math, so ____.** *División decimal importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about decimal division.
Di un hecho verdadero sobre decimal division.

Decimal division ____.

Question
Pregunta

Ask a question about decimal division.
Haz una pregunta sobre decimal division.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about decimal division.
Muestra entusiasmo sobre decimal division.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with decimal division.
Dile a un compañero qué hacer con decimal division.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I make the divisor 6.3 into a ____ before dividing.
Convierto el divisor 6.3 en un ____ antes de dividir.

Dividing by a decimal directly is hard because ____.
Dividir entre un decimal directamente es difícil porque ____.

I moved both decimal points ____ place, changing it to ____ ÷ ____.
Moví ambos puntos decimales ____ lugar, cambiándolo a ____ ÷ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is 15.6 ÷ 1.3?

A. 12

B. 1.2

C. 120

D. 13

Show your work:

2. A ribbon is 16.8 feet long. You cut it into pieces that are each 2.4 feet long. How many pieces do you get?

A. 7 pieces

B. 8 pieces

C. 6 pieces

D. 14.4 pieces

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A gas station sells fuel at $3.45 per gallon. A customer pays $27.60 total. How many gallons did the customer buy? Set up the decimal division, show how to convert it to whole-number division, and solve.

Sentence starter: The equation is $27.60 ÷ $3.45 = ____. I multiply both by ____ to get ____ ÷ ____ = ____ gallons.

Show your work:

____

____

____

____

Reflect — Exit Ticket

What is 19.2 ÷ 0.8?

A. 24

B. 2.4

C. 240

D. 0.24

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. 12 *— Multiply both by 10: 156 ÷ 13 = 12.*
2. **Try It 2:** A. 7 pieces *— 16.8 ÷ 2.4: multiply both by 10 → 168 ÷ 24 = 7 pieces.*
3. **Exit Ticket:** A. 24 *— Multiply both by 10: 192 ÷ 8 = 24.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Decimal division is dividing with decimals. First make the number you divide by a whole number.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).